MICRO PRECISION

MICRO PRECISION CALIBRATION INC.
9080 ACTIVITY RD. STE C
SAN DIEGO CA 92126
858-547-0217

Certificate of Calibration

Date: Jan 3, 2025 **Cert No.** 5523631031482435

Customer:

FUERZA AEREA DE CHILE
RODRIGO DE ARAYA # 1263
SANTIAGO CHILE 7810161

MPC Control #:	31931
Asset ID:	31931
Gage Type:	TRANSDUCER VIBRATION VELOCITY
Manufacturer:	CEC
Model Number:	09384/4-123-0001
Size:	45 to 2000 Hz
Temp/RH:	22.2°C / 46.8%
Location:	Calibration performed at MPC facility

Work Order #:	SD-50051345
Purchase Order #:	PO-58582-569
Serial Number:	31931
Department:	N/A
Performed By:	MIGUEL MATA
Received Condition:	IN TOLERANCE
Returned Condition:	IN TOLERANCE
Cal. Date:	October 30, 2024
Cal. Interval:	12 MONTHS
Cal. Due Date:	October 30, 2025

Calibration Notes:

Instrument calibrated in accordance with manufacturer's specifications.
See attached calibration data (1 page).
Procedure used as reference.

This report replaces obsolete report number 5523631031340086
Company address was updated.

Standards Used to Calibrate Equipment

I.D.	Description.	Model	Serial	Manufacturer	Cal. Due Date	Traceability #
P1630	TRANSDUCER TEST SET	HI 803	0355	HARDY INSTRUMENTS	Jul 31, 2025	5523631031288279
DE5574	DIGITAL MULTIMETER	3458A	2823A14502	HEWLETT PACKARD	May 31, 2025	5523631030922025
BQ4564	DECADE RESISTOR	1432-L	31612	GENERAL RADIO COMPANY	Jul 31, 2025	5523631031055789

Calibrating Technician: MIGUEL MATA

QC Approval: JESSE GARCIA

STATEMENTS OF PASS OR FAIL CONFORMANCE: The uncertainty of measurement has been taken into account when determining compliance with specification. All measurements and test results guard banded to ensure the probability of false-accept does not exceed 2% in compliance with ANSI/NCSL Z540.3-2006.

THE CALIBRATION REPORT STATUS:
PASS- Term used when compliance statement is given, and the measurement result is PASS.
PASS²- Term used when compliance statement is given, and the measurement result is conditional passed or PASS².
FAIL- Term used when compliance statement is given, and the measurement result is FAIL.
FAIL²- Term used when compliance statement is given, and the measurement result is conditional failed or FAIL².
REPORT OF VALUE - Term used when reported measurement is not requiring compliance statement in report.
ADJUSTED- When adjustments are made to an instrument which changes the value of measurement from what was measured as found to new value as left.
LIMITED - When an instrument fails calibration but is still functional in a limited manner.

The expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k=2, which for a normal distribution corresponds to a coverage probability of approximately 95%, unless otherwise stated.
This calibration report complies with ISO/IEC 17025:2017, ANSI/NCSL Z540.3-2006 and ANSI/NCSL Z540.1-1994. Calibration cycles and resulting due dates were submitted/approved by the customer. Any number of factors may cause an instrument to drift out of tolerance before the next scheduled calibration. Recalibration cycles should be based on frequency of use, environmental conditions and customer's established systematic accuracy. All standards are traceable to SI through the National Institute of Standards and Technology (NIST) and/or recognized national or international standards laboratories. Services rendered include proper manufacturer's service instruction and are warranted for no less than thirty (30) days. The information on this report pertains only to the instrument identified, this may not be reproduced in part or in a whole without the prior written approval of the issuing MP Calibration Laboratory.

(CERT, Rev 9)

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Certificate of Calibration

Date: Jan 3, 2025

Cert No. 5523631031482435

Procedures Used in this Event

Procedure Name	Description
MPC-VIB-001 Rev. 06	Accelerometers, General, Rev.06, Jun-07-2024

Calibrating Technician:

MIGUEL MATA

QC Approval:

JESSE GARCIA

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PASS^{z}- Term used when compliance statement is given, and the measurement result is conditional passed or PASS^{z}.
FAIL- Term used when compliance statement is given, and the measurement result is FAIL.
FAIL^{z}- Term used when compliance statement is given, and the measurement result is conditional failed or FAIL^{z}.
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MICRO PRECISION

Calibration Report of CEC 09384/4-123-0001 Transducer Vibration Velocity

MPC Control #:	31931	Serial Number:	31931
Asset ID:	31931	Calibration Date:	October 30, 2024

Acceleration Measurement

Reference Sensitivity / Mass 112.462 grams

Applied Setting	UUT Spec Sensitivity	Lower Limit	As Found	As Left	Upper Limit	Result
100 Hz @ 1 in/sec	135.0 mV	133.0000 mV	134.2537 mV	134.2537 mV	137.0000 mV	PASS

Frequency Response

Referenced to 100 Hz Output

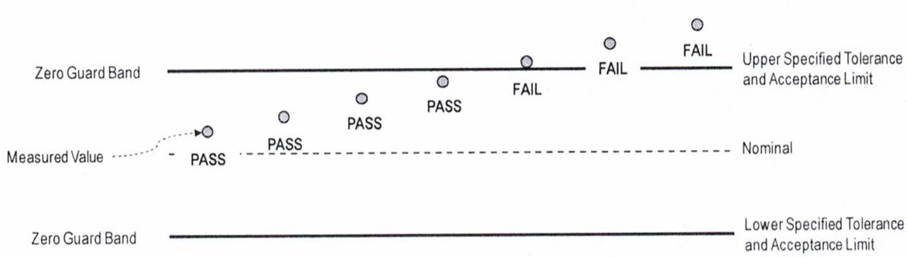
Applied Setting	UUT Ref Sensitivity	Lower Limit	As Found	As Left	Upper Limit	Result
45 Hz @ 1 in/sec	134.2537 mV	123.5134 mV	134.2425 mV	134.2425 mV	144.9940 mV	PASS
1000 Hz @ 1 in/sec	134.2537 mV	123.5134 mV	135.3361 mV	135.3361 mV	144.9940 mV	PASS
1500 Hz @ 1 in/sec	134.2537 mV	123.5134 mV	135.7222 mV	135.7222 mV	144.9940 mV	PASS
2000 Hz @ 1 in/sec	134.2537 mV	123.5134 mV	136.0899 mV	136.0899 mV	144.9940 mV	PASS

Statements of Pass or Fail Conformance

The status of compliance with the acceptance criteria is reported as:

PASS — Compliant with specification.

FAIL — Not compliant with specification.



Acceptance limits equal to the tolerance limits

- End of Calibration Report -